

CHAPTER 1 SUMMARY, CONCLUSION, AND FUTURE WORK

1.1 Summary of Findings

This study developed and evaluated FeatureSAGE, a modification of the SAGE framework that integrates the StyleFeatureEditor inverter and Feature Editor into the few-shot attribute group editing pipeline. The findings are summarized according to the main contributions and evaluation objectives of the study.

1. Integration of the StyleFeatureEditor inverter. FeatureSAGE replaces the pSp-centered inversion stage of SAGE with the StyleFeatureEditor inverter, allowing the framework to represent input images using both W^+ latent codes and intermediate feature tensors. This modification produced lower FID scores than both baselines on the two evaluated datasets. At 1024×1024 resolution, FeatureSAGE obtained an FID of 45.60 on 102 Flowers and 49.60 on Animal Faces.

2. Use of feature-level editing. The Feature Editor modifies intermediate feature representations during the decoder stage, supporting detail preservation while latent-space edits control semantic variation. Qualitative comparisons show sharper petal boundaries, clearer flower structure, and more coherent animal-face texture and shading in the FeatureSAGE outputs compared with the displayed AGE and SAGE outputs.

3. Quantitative improvement over AGE and SAGE. On the 102 Flowers dataset, FeatureSAGE improved FID by 15.8% over SAGE and 25.9% over AGE. On the Animal Faces dataset, FeatureSAGE improved FID by 18.6% over SAGE and 20.2% over AGE. These results show that the proposed inversion and feature-editing changes improved distributional alignment with real images under the one-shot setting.

4. Preservation of perceptual variation. FeatureSAGE maintained LPIPS values close to the baselines while achieving the best FID scores. On 102 Flowers, FeatureSAGE obtained an LPIPS of 0.4209, compared with 0.4528 for SAGE and 0.4108 for AGE. On Animal Faces, FeatureSAGE obtained an LPIPS of 0.4811, compared with 0.4867 for SAGE and 0.5033 for AGE. Thus, the improvement in FID did not remove perceptual variation among generated samples.

1.2 Limitations

The findings are constrained by the scope of the study. First, FeatureSAGE depends on a pretrained StyleGAN2 generator, so its applicability is limited to domains where an appropriate generator is available and sufficiently representative of the target categories. Any coverage gaps or biases in the pretrained generator can influence the generated outputs.

Second, integrating the StyleFeatureEditor inverter and Feature Editor increases computational requirements relative to the baseline SAGE pipeline. The additional feature extraction, feature editing, and generator passes improve reconstruction and output quality, but they also increase GPU memory use and processing time.

Third, the quantitative evaluation is limited to the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting at 1024×1024 resolution. The conclusions should therefore be interpreted within these datasets, metrics, and experimental conditions.

1.3 Future Work

Future research may focus on reducing the computational overhead introduced by the StyleFeatureEditor inverter and Feature Editor. More efficient inversion or feature-editing modules could preserve the FID improvements reported in this study while lowering GPU memory use and inference time.

Another direction is to evaluate FeatureSAGE with generator backbones beyond StyleGAN2. Since the present study is limited to the StyleGAN2-based SAGE pipeline, testing

other generator architectures would clarify whether the same feature-level editing strategy generalizes beyond the current framework.

Future work may also examine multi-level feature editing. The current implementation uses an intermediate feature tensor for feature injection; extending the approach to multiple feature levels could further test whether local detail preservation and semantic consistency can be improved without sacrificing perceptual variation.

1.4 Conclusion

This study addressed the problem of improving few-shot image generation in the SAGE framework, particularly the limitation caused by relying on pSp-based inversion. The proposed FeatureSAGE framework integrates the StyleFeatureEditor inverter and Feature Editor so that images are represented not only by W^+ latent codes but also by intermediate feature tensors used during synthesis.

The results answer the research problem affirmatively within the scope of the evaluated datasets and metrics. Integrating the StyleFeatureEditor inverter and Feature Editor improved generated-image distribution alignment under the one-shot setting. FeatureSAGE achieved the lowest FID in all reported comparisons: 45.60 on 102 Flowers and 49.60 on Animal Faces. These correspond to FID improvements of 15.8% and 18.6% over SAGE, and 25.9% and 20.2% over AGE, respectively. At the same time, LPIPS remained comparable to the baselines, with 0.4209 on Flowers and 0.4811 on Animal Faces, showing that the fidelity gains did not eliminate perceptual variation.

The findings address the research questions as follows. First, the integration of the StyleFeatureEditor inverter and Feature Editor improved the inversion-and-editing pipeline as reflected by lower FID scores and sharper qualitative outputs. Second, the modifications influenced generated-image quality while maintaining comparable LPIPS values, indicating that consistency and diversity remained balanced in the reported one-shot setting. Third, FeatureSAGE retained the SAGE attribute group editing structure while adding feature-level

editing, supporting reliable category-preserving edits in the displayed qualitative results. Fourth, compared with AGE and SAGE, FeatureSAGE achieved the strongest overall quantitative performance on FID across both datasets while remaining close to the baselines in LPIPS.

The study contributes four main findings: it integrates StyleFeatureEditor inversion into SAGE, adds feature-level editing through the Feature Editor, demonstrates consistent FID improvement over AGE and SAGE on both datasets, and shows through LPIPS and qualitative comparisons that perceptual variation and visual coherence are maintained. Therefore, FeatureSAGE provides a stronger few-shot attribute group editing framework for the tested Animal Faces and 102 Flowers settings, while its dependence on StyleGAN2 and its higher computational cost remain important limitations for future work.